QUIZSENSE: A WEB APPLICATION FOR MULTIPLE-CHOICE QUESTION QUIZ MAKER

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BACHELOR OF SCIENCE IN COMPUTER SCIENCE

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INTRODUCTION

Background of the Study

Technology has transformed numerous industries, and education is no exception. Education is the process of acquiring knowledge, skills, values, and habits through various means such as teaching, training, and experience, essential for intellectual, social, and emotional growth. It takes place in formal settings like schools, informal environments like homes, and non-formal program such as workshops. The primary goals of education are to foster critical thinking, creativity, and problem-solving, build foundational knowledge in key areas, and promote ethical and cultural awareness. It also develops life skills like communication, teamwork, and adaptability, preparing individuals for personal and professional success while enabling them to contribute to society.

According to Mdhlalose and Mlambo (2023), technology has become an integral part of education. Virtual learning environments, online courses, and educational software have made learning accessible to anyone with an internet connection. Innovations such as artificial intelligence (AI) and augmented reality are transforming the way students learn by making education more personalized, engaging, and effective. The integration of advanced technologies and learning platforms has opened up a world of opportunities, enabling more interactive and efficient educational experiences.

According to Khairkar et al. (2024), both teachers and students find the standard exam creation procedure to be laborious and sluggish, which causes a delay in the announcement of results. It is widely acknowledged that the most effective method of evaluating students' comprehension is through multiple-choice questions (MCQs). With the help of a quiz generator bot, exam creation bot, interactive quiz bot, and grading bot, this project aims to create objective questions with accurate answers based on user-inputted text. By streamlining the entire process, it will help create assessments that are both challenging and interesting for students. A multiple choice quiz significantly benefits students by providing an engaging, efficient, and effective way to learn and assess their knowledge. It allows students to reinforce their understanding through regular practice, helping them retain concepts better and encouraging active recall, which strengthens memory.

According to Gabajiwala et al. (2022), deep learning and natural language processing have seen significant technological advancements in recent years. Making quizzes automatically from text is one of the uses. Recent developments in natural language processing algorithms have showed great promise.

QuizSense is a web-based application that aims to help students with their subjects, specifically in the fundamentals of programming. A web-based tool called QuizSense will assist students with their coursework, particularly in fundamentals of programming topics in each chapter. To improve the overall

learning experience, it will be possible to automate the creation of relevant, excellent quiz questions through the use of machine learning, file parsing, and OCR processing techniques.

This proposal aims to leverage machine learning, file parsing, and OCR processing techniques to generate quizzes and suggest topics. A web application is expected to reduce the workload of creating quiz questions and can be used for trial tests, as well as to identify subjects that require improvement based on students' quiz results.

Significance of the Study

The significance of the study lies in developing a web application designed for creating multiple-choice quiz that offers an educational assessment and learning outcomes. This research aims to provide value to various stakeholders:

1. **For Students**: This study will benefits students by allowing them to engage in self-assessment through multiple-choice quizzes. Students can reinforce their learning, test their knowledge, and identify their weak areas.
2. **For Educators:** This study offers educators a valuable resource by simplifying the process of quiz creation. This helps educators better understand students' performance and refine their teaching strategies.

3. **For Academic Researchers:** This research contributes to educational technology by demonstrating the practical use of web applications in improving learning outcomes.

Objective of the Study

The main objective of the study is to create a web application that can generate quiz questions and multiple-choice answer base on the 5 chapters of Fundamentals of Programming with OCR and File Parsing algorithm.

Specifically, the study aims to:

1. gather pdf or word file in lecture of fundamentals of programming as dataset;
2. evaluate the accuracy of the model using confusion matrix;
3. develop a web application that allows students to take a quiz;
4. recommend topics improvement based on quiz performance; and
5. evaluate the usability and functionality of the web application.

Expected Output of the Study

The expected output of the study is a created web application that generates quiz based on pdf and word file topics from the 5 chapters of Fundamentals of Programming, including Introduction to Computer Programming, Programming Fundamentals, Inputs and Outputs, Control Structures, and Arrays. The application will also analyze quiz results to recommend areas for improvement.

Scope and Limitation of the Study

This study focuses on developing a web application that generates quizzes based on five chapters covering the fundamentals of programming, including Introduction to Computer Programming, Programming Fundamentals, Inputs and Outputs, Control Structures, and Arrays.

The web application will support the export of quizzes in PDF and Word document file formats. Each quiz generated will contain 10 questions, with 4 choices for each question. The datasets for generating quizzes and answers will be gathered from internet sources, specifically Google Scholar and Google Chrome websites.

Additionally, students will evaluate the usability, efficiency, and overall functionality of the web application. This research involves utilizing both training and testing datasets to generate quizzes and multiple-choice answers.

Furthermore, the web application will provide detailed feedback on areas within the fundamentals of programming that require improvement based on quiz results obtained from users or students. The Optical Character Recognition (OCR) algorithm will be implemented solely for recognizing and extracting text from documents, while file parsing techniques will be applied to process and structure the extracted data effectively.

Operational Definition of Terms

Terms are operationally described below for the readers' benefit.

Accuracy – is a phrase that is used to describe the qualities of accuracy.

Algorithm – is a technique that uses specific instructions to calculate a property represented as a finite collection of directions. An estimation that starts with a starting input and state, progresses with a starting input and state of clearly defined state, generates output, and concludes in a final state is stated when the ordering is completed.

Application - is software or a program created to carry out particular duties for users. Applications in computers are made to entertain, solve issues, or boost efficiency and productivity.

IDE - when developing software, a programmer uses a development environment that is integrated, which there is a program with an edit source code, integrated tools, as well as a debugger.

Fundamentals of Programming – programming fundamentals refer to the core principals and building blocks that underpin all software development. These include concepts like variables, data types, control structures, functions, and algorithms.

File parsing – is use to utilized to read and decipher a file's contents, converting unstructured data into a format that a program can process and analyze with ease. Applications like data extraction, configuration file

interpretation, and programming language compilation all depend on parsing.

Generate Quiz – is the process of developing a test or series of questions intended to evaluate a subject's knowledge or comprehension. Either physical labor or automated tools and software can be used for this.

Optical Character Recognition – is a technique that transforms a variety of document types including scanned paper documents, PDF files, and camera images into data that can be edited and searched. Acquisition of images, preprocessing to improve quality, text identification, character recognition, and post-processing to fix errors are all processes in the OCR process.

Programming Language – is a computer programming language that enables to programmers symbolically exemplifies data processing without worrying about machine details.

Conceptual Framework

Figure 1 shows the study begins with data collection, processing and file parsing the datasets to ensure quality. Pre-processing prepares the data for analysis, which includes tokenization to break it down into smaller units and feature extraction for training the model. The data is then split into training and testing. An Optical Character Recognition (OCR) algorithm will be employed to recognize the text from the collected data, evaluate the training and testing datasets using confusion matrix enabling the system to generate quizzes based on the processed information. Users can take the generated quizzes, and their performance will be evaluated to produce a score. Based on the results, the web application recommends topics for improvement, and the overall usability and functionality of the system are evaluated by the users.

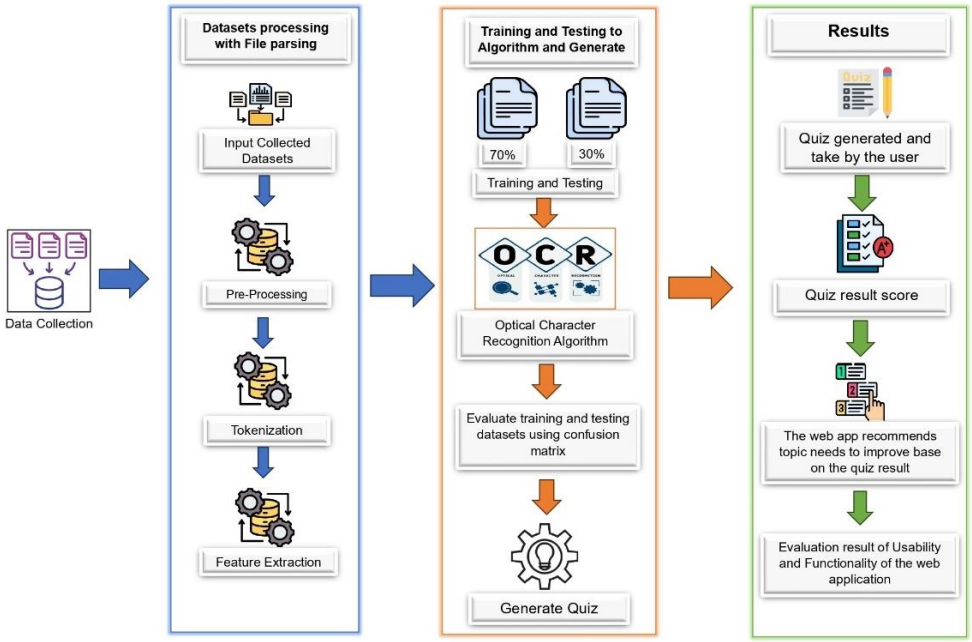

Figure 1. Conceptual Framework of the Study

REVIEW OF RELATED LITERATURE

This section covers the literature reviews that were relevant to the methods and procedures used in this study.

Quiz Maker

Hedström et al. (2021), created a quiz on personality that uses machine learning and the scientific method to identify which character from the television show Friends the test taker most closely resembles. A split training/test dataset was created by extracting features from each episode's manuscript. Then using the test data and performance metrics like Akaike information criterion, k-fold cross-validation, and principal component analysis, various models from the Python modules SKLearn and Pytorch, including neural networks, quadratic discriminant analysis, and decision trees, were trained and assessed. While gradient boosting yielded the highest accuracy at 78%, because it made it simpler to determine which factors were significant to which character by examining the model's coefficients, logistic regression—which had an accuracy of 59% following k-fold cross validation—was selected as the approach to use as the foundation for the test.

According to Shi et al. (2021), since smartphones and tablets have become so widespread, e-learning has advanced significantly. Learners are being given access to increasingly sophisticated e-learning resources due to

advancements in device performance and internet data transmission capabilities. The platform utilizes linked data to store both values and relational data, enabling the construction of extensible and modifiable quiz systems. This approach facilitates the generation of quizzes that can be readily adapted and enhanced, while also providing learning analytics derived from user activity. To augment educational efficacy, the framework is extended to support quiz creation within 3D environments. Specifically, the application of this framework within the domain of medical education is explored.

In a study conducted by Gupta et al. (2021), they conducted multiple-choice questions based on any text image, whether from lengthy, dull books or even your own handwritten notes. The system aggregates questions from diverse online sources, inclusive of text-based images, to maintain a current and comprehensive question database. This approach enables the inclusion of contemporary information and facilitates ongoing review. An automated quiz generation system is implemented through the integration of multiple natural language processing (NLP) methodologies. Users input text image data via a provided API, which then utilizes web-based search functionalities to produce multiple-choice questions (MCQs). The application of these NLP techniques aims to automate and optimize the quiz creation process.

Khairkar et al. (2024), both teachers and students find the traditional exam creation process to be time-consuming and inefficient, leading to delays in the release of results. Multiple-choice questions (MCQs) are widely

recognized as one of the most effective methods for assessing students' comprehension. To address these challenges, a quiz generator bot, exam creation bot, interactive quiz bot, and grading bot are utilized to generate objective questions with accurate answers based on user-inputted text. By streamlining the entire assessment process, this approach facilitates the creation of quizzes that are both engaging and intellectually stimulating for students. Utilizing NLP pre-processing techniques, WordNet, PKE, the T5 Algorithm, and the BERT Algorithm, the platform generates quizzes from randomly provided text in formats such as PDF, DocX, or TXT, ensuring efficiency and accuracy in exam creation.

Gabajiwala et al. (2022), deep learning and natural language processing have seen significant technological advancements in recent years. Making quizzes automatically from text is one of the uses. Recent developments in natural language processing algorithms have showed great promise. In order to extract keywords and obtain insights from the text input, the suggested solution employs an NLP pipeline that includes Bert and T5 transformers.

Wang et al. (2023) introduced iQS, an AI-assisted quiz system aimed at supporting self-regulated and remote learning. This system can generate individual tests and deliver immediate, personalized feedback. iQS is compatible with various learning management systems (LMS), including Moodle and MOOCs, making it versatile and easily implementable. Its core algorithms are reusable, requiring only specialized domain knowledge and learners' data

to function in a new context. Developed and tested within a university's Moodle platform, iQS received positive feedback from both teachers and students. The system addresses the need for personalized learning, self-assessment, and a scalable AI-powered quiz platform. Feedback from iQS significantly enhances students' cognitive engagement, satisfaction, motivation, and interaction with the learning material, demonstrating its impact and effectiveness.

Optical Character Recognition Algorithm

According to Arica and Yarman-Vural (2022), optical character recognition for cursive off-line cursive handwriting recognition is approached with a novel analytical strategy that employs a series of image segmentation and recognition techniques. First, some global parameters are estimated, including baselines, height, stroke width, and slant angle. Second, a segmentation technique combines binary and grayscale data to determine character segmentation pathways. Third, the character candidates are labeled and ranked using a hidden Markov model (HMM) for shape identification. The charac ter candidates are represented by a string of codes that are taken from each segment for this purpose, for word-level recognition, a graph optimization problem combines data from a lexicon and the HMM ranks.

Chaudhuri et al. (2017), the process of classifying the optical patterns present in a digital image is known as optical character recognition (OCR).

Segmentation, feature extraction, and classification are used to recognize the characters. Optical scanning, location segmentation, pre-processing, segmentation, representation, feature extraction, training and recognition, and post-processing are some of the several OCR system methodologies. The various uses of OCR systems are then discussed, and finally, the state of OCR systems today. The OCR systems' future is finally discussed.

Patel et al. (2016) discussed the application of Optical Character Recognition (OCR) techniques in converting printed text into editable digital text. OCR is a highly useful and widely adopted technique across various applications. The accuracy of OCR largely depends on text segmentation and preprocessing algorithms. However, challenges arise in extracting text from images due to variations in font size, style, orientation, and the complexity of the image background. Among the OCR tools, Tesseract is noted for its superior accuracy, while Transym OCR is recognized as one of the proprietary OCR solutions. The effectiveness of these tools highlights the importance of selecting appropriate algorithms and preprocessing techniques to achieve optimal OCR performance.

Hamad and Kaya (2016), the goal of optical character recognition research is to create a computer system that can automatically extract and process text from photographs OCR's goal is to modify or convert any type of text or documents that contain text, including handwritten writing text images that have been printed or scanned and converted into an editable digital format

for additional editing. OCR thus makes it possible for a machine to automatically identify text in these kinds of documents. For automation to be successful, a few key obstacles must be identified and addressed. The clarity of photos and the font features of the characters in paper documents are just two examples of the recent difficulties. These difficulties can occasionally cause characters to be incorrectly detected by computer systems.

Shastry et al. (2016), in the history of electronics and image processing, research in the fields of computer vision, artificial intelligence, and pattern recognition has long been significant. One of the primary facets of computer vision is optical character recognition (OCR), which has advanced significantly since its debut. OCR is a technique that uses digitally acquired optical data to identify readable characters. Various algorithms and techniques have been developed to enhance its accuracy and efficiency. Among these, a method referred to as "i" is introduced as a fast, user-friendly OCR system that operates independently of font and size constraints. This system utilizes a unique segment extraction method, allowing it to identify individual alphabetic characters without relying on complex mathematical computations. Designed to serve as a core component within a broader OCR solution, this approach enhances text recognition capabilities while maintaining simplicity and efficiency.

Generate quiz with OCR

According to Dulay et al. (2016), Wi-Fi-based data transport system for applications that use automated engineering quiz contests. Each school participant's "answer sheet" is an Android app that is part of the system. Optical Character Recognition (OCR) is the recognition system in use. The system simulates the standard engineering quiz contest, in which participants are given a set amount of time to respond to questions that flash on the screen. In order to assess the network's ability to process the broadcast signal at the same time, the system was tested using fifteen devices operated by fifteen individuals. A grace period is given to allow all of the images to pass through, and the timer that flashes on the screen is synchronized with the image-to-text converter's timer. The accuracy of the OCR's character recognition is 83.95%. For responses that contain many letters and symbols, the detection accuracy is only dependent on the lettering style. The image is transformed erroneously when a letter overlaps another letter because the OCR sees it as a different character.

Sreelakshmi et al. (2019), numerous chatbots have been created in the subject of education in recent years using ORC algorithm. While these chatbots are designed to answer queries based on publicly available information, such as data from community response platforms or pre-existing knowledge bases, they often lack the ability to customize the information being queried. To address this limitation, a Quiz Generation Chatbot has been developed to facilitate both question generation and answer extraction based on user-inputted documents.

The system undergoes several data cleaning and preprocessing procedures to transform the uploaded content into a structured knowledge base. The Quiz Generation Module identifies key sentences and formulates question-answer pairs, which can then be used to generate quizzes for users. Meanwhile, the Question Answering Module utilizes neural networks and ranking functions to extract the most relevant responses from the knowledge base. By integrating these functionalities, the chatbot enhances educational engagement by allowing users to create customized quizzes while ensuring accurate and contextually relevant answers.

Aizawa and Korpi (2016), as optical character recognition methods advance, new avenues for enhancing current systems become available. For instance, there are apps for interpreting foreign language by taking pictures with cellphones and OCR reading aids for the blind. But the learning environment is one area that hasn't benefited from the new technologies. The main focus of this demonstration is adding optical character recognition to a smartphone app that teaches Japanese kanji characters. Users can use the app to snap pictures of kanji in their daily surroundings and quickly find extensive information and translations. With the use of vocabulary lists and quiz, users can practice those kanji and monitor their progress.

Sornavalli et al. (2023) introduce a digital learning platform designed to cater to diverse student needs, emphasizing a dynamic approach to education. Key features, including a recommendation algorithm tailored to individual

learning styles, text summarization for efficient information synthesis, adaptive quiz generation, and the integration of Optical Character Recognition (OCR) with image processing for handwritten text conversion, work in concert with scheduling algorithms to enhance time management. Driven by the ambition to revolutionize education, the authors combine these technologies, employing Natural Language Processing (NLP) and Machine Learning (ML), to create a personalized learning experience. They recognize that traditional teaching methods often struggle to accommodate the unique learning preferences and styles of each student, highlighting the necessity for an innovative educational platform that addresses these limitations and provides a more tailored learning journey.

Eftimie et al. (2016), an essential component of education is student testing. Classic paper quiz examination is still the favored mode of examination because of its minimal technical requirements and ease of execution, despite the development and popularity of different electronic testing and grading solutions in recent years. With inline answer markings, answer sheet processing via auto feed scanners and OCR detection, and automatic test production, processing, and grading, Korect offers a comprehensive solution. The research done to optimize the pretreatment stage and the scan folder polling during the automatic verification flow is presented in this project.

Fundamentals of Programming

According to Oyarvide et al. (2023), understanding the fundamentals of programming, particularly the use of selective and repeating structures (loops), is often too intricate and abstract for students. To address this, the study suggests the need for intuitive and useful visualization techniques, as programming structured applications requires a clear understanding of the relationship between variables, operators, and declarations. By employing dynamic and interactive flowcharts, college students pursuing entry-level computer science majors can learn how to program structured applications through visual representation and interaction. The study involved 34 first-year university students from Guayaquil, who, despite passing a 16-week "fundamentals of programming" course, struggled with developing programming logic. The results recommended adopting a framework comprising cognitive, motivational, usability, and affective aspects to achieve the right balance of usability and enjoyment, enabling students to learn efficiently and enjoyably. The study also compared the use of traditional paper flowcharts with augmented reality-enhanced versions, demonstrating the potential benefits of advanced visualization techniques in improving programming education.

According to Insuasti et al. (2023), an approach, a model, and an adapted theory have been developed for teaching key programming concepts such as variables, expressions, comparisons, loops, and functions in computer science. The primary contribution of this study is the expansion of the concept of didactic

transposition to enhance context-based instruction. The concept is known as didactic transposition. According to the proposal, a linguistic corpus based on curricula, textbooks, reference guides, and a survey given to professionals who teach programming basics at seventy-eight universities across the globe will be developed. A modified theory, an approach, and a model for teaching the foundations of programming through computational linguistics are created. Furthermore, a template for developing instructional strategies in this area is developed. A number of factors are taken into consideration when developing teaching strategies using computational linguistic techniques from written documents, and the proposal offers a significant alternative for doing so where the context's conditions are prioritized..

Medveckis et al. (2021), highlight lifelong learning as a crucial paradigm of the 21st century, emphasizing the continuous development of professional competence. Their study evaluates the effectiveness and quality of the "Fundamentals of Programming in Visual Programming Environment Scratch" program, which aims to enhance educators' professional skills. The evaluation criteria include program structure, presenter expertise, and quality of materials, content relevance, expectations, practical application, and overall usefulness. The study, conducted among 98 Latvian instructors, used a combination of open-ended and closed-ended questions for data collection, with SPSS 20 employed for statistical analysis. Results analyzed through correlation coefficients indicate that educators who participated in the program significantly

improved their programming knowledge and digital competency. This study is relevant to the fundamentals of programming as it underscores the importance of structured programming education, particularly in visual programming environments like Scratch. By demonstrating how professional development programs can enhance both programming proficiency and digital literacy, the research reinforces the value of integrating foundational programming education into teacher training, ultimately contributing to more effective programming instruction for students.

Luxton-Reilly et al. (2018), highlight the broad range of activities included in introductory programming courses, such as creating, implementing, testing, and debugging programs that apply fundamental programming principles. These activities shape expectations, assessments, and instructional approaches within programming education. While research has demonstrated the value of assessing single concepts, most formal examinations integrate multiple programming concepts, increasing complexity and making evaluations more challenging. This complexity often leads to inadequate feedback for students regarding their progress and difficulties, while instructors struggle to pinpoint the specific challenges students encounter. The limitations in feedback quality may contribute to high dropout rates and the widespread perception that introductory programming is difficult. Researchers in computer science education emphasize the importance of understanding the syntax and semantics necessary for programming assessments, as these are critical for

novice programmers. By surveying the desired learning outcomes of these courses, they analyze the relationships between different programming elements and the development of fundamental concepts. This study is relevant to the fundamentals of programming as it underscores the importance of structured assessments, targeted feedback, and concept-based learning, all of which play a crucial role in ensuring students grasp essential programming principles effectively.

Boiarinova et al. (2023), emphasize that the fundamentals of programming are a core component of first-year science and engineering courses. To enhance students' understanding of how coding can be applied in their studies and research, the authors have integrated Python programming with real-world applications in science and engineering, along with research-oriented skills, into a newly developed first-year curriculum. The initiative has received positive feedback from students, and ongoing refinements are being made based on faculty and student input. As course enrollment grows, it is expected that increased computational literacy will encourage more undergraduate and graduate students to adopt computational methods in their academic and professional pursuits. This study is highly relevant to the fundamentals of programming as it highlights the importance of integrating programming education with practical applications. By demonstrating how coding skills can be applied across disciplines, the study reinforces the significance of foundational programming knowledge in preparing students for

problem-solving and computational thinking, which are essential skills in modern scientific and engineering fields.

METHODOLOGY

This section provides an overview of the procedures and techniques applied in this study. It details the process for data collection, preparation, and analysis, as well as the tools and algorithms used.

Research Design

This study will use both experimental and descriptive researcher design and method, Figure 2 shows the block diagram of the entire system and describes in general terms of the web application quiz maker. It starting with a User Inputs Collected Data, Pre-processing, Tokenization, Feature Extraction and File parsing is applied on the process then after feature data which is the training and testing data proceed to OCR algorithm, generate quiz and result of the quiz taken by the user and the web application gives a recommendation topic needs to improve by the user base on the quiz result.

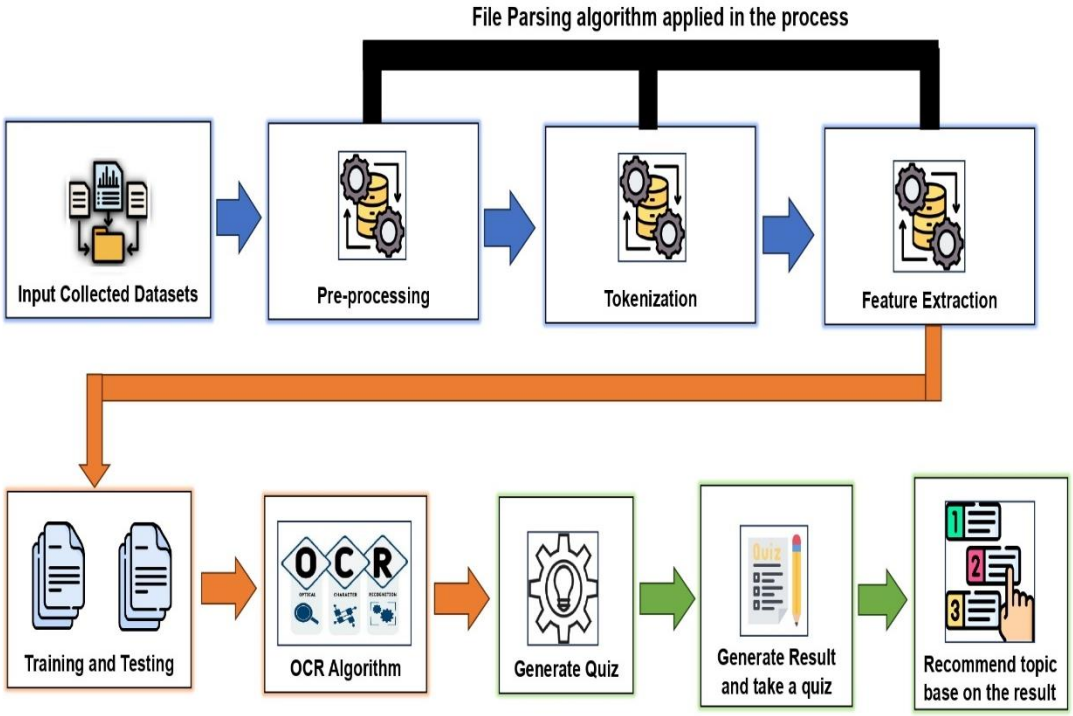

Figure 2. Block Diagram of Research Design

Role of the Researcher

In this study, the researcher aims to develop a web application to generate multiple-choice quizzes for students. The researcher will be responsible for creating a framework for the web application. Additionally, the researcher will perform a test on 30 students under DCLIS, using a questionnaire to evaluate the usability and functionality of the web application.

Research Instrument

The primary research instruments for this study include the developed multiple-choice quiz maker using OCR and file parsing algorithms. The developed web application will process the uploaded dataset, which consists of thirty-eight (38) topics from five chapters of Fundamentals of Programming, and will be ready for assessment.

Data Collection

The researcher will gather thirty-eight topics from five chapters covering the fundamentals of programming, including Introduction to Computer Programming, Programming Fundamentals, Inputs and Outputs, Control Structures, and Arrays, from the internet, specifically using Google Scholar and Google Chrome, to create a new dataset. The total of 1,000 collected files will be divided into two sets: 70% (700 files) will be used as training datasets, while

30% (300 files) will be used as testing datasets. The training dataset will consist of 700 files, with each chapter containing 140 files since the dataset is evenly divided among the five chapters. Similarly, the testing dataset will consist of 300 files, with each chapter containing 60 files. The web application will examine these testing files to generate quizzes and suggest topics for improvement based on the user's quiz results.

Place and Time of the Study

The research takes place at the University of Southern Mindanao – Main Campus from December 2024 to April 2025.

Target sample list of data

Table 1 shows the target list of topics per chapter to collect by the researcher as datasets.

Table 1. List of Topics by Chapter

Chapter 1	Chapter 2	Chapter 3	Chapter 4	Chapter 5
What is Computer Programming?	Program Development Life Cycle	Getting to Know Software Interface	Design Control Structure	Introduction to Arrays
Brief History of Programming	Comments	Creating Simple Programs	If	Declaring Arrays
Why Learning Programming?	Tokens	Displaying Outputs on Console	If-else	Accessing an Array Element
Traits of a Good Programmer	Separators	Getting Input and Displaying Output using Graphical User Interface	If-else-if	Array Length
Good Programming Practices	Identifiers	—	switch	Multidimensional Arrays
Qualities of a Good Program	Keywords	—	Repetition Control Structure	—
—	Literals	—	While	—
—	Data Types	—	do-while	—
—	Variables	—	For	—
—	Operators	—	—	—
—	Expressions	—	—	—
—	Statements	—	—	—
—	Blocks	—	—	—

Validity

The collected dataset by the researcher will be validated by an expert in the field, whether the dataset is Chapter 1, 2, 3, 4, 5 topics in Fundamentals of Programming. The expert will also supervise the dataset used for training and testing to ensure that the datasets are not manipulated and the study is not a biased.

Ethical Consideration

To guarantee that participants are respected and protected from any kind of harm physical, psychological, or emotional the researcher will closely adhere to ethical standards. Secure data storage and restricted access to personal data will protect participants' privacy and guarantee study confidentiality. Participants will get clear information about the objectives, methods, and potential risks and benefits of the research through open, truthful, and transparent communication. The study process will become respectful and accountable as a result of this dedication to ethical values, which will lay the groundwork for trust and honesty.

Identification of the Problem

The suggested study is focused on the creation of an approachable web application that can generate quiz of fundamentals of programming topic and give a suggestion topic needs to improve base on the result of the quiz.

Analysis of the Problem

The study will use the following:

I. Input Requirements

Collected fundamentals of programming topic files by chapter.

II. Output Requirement

The application will show can generate quiz of fundamentals of programming topic and give a suggestion topic needs to improve base on the result of the quiz.

Data Analysis

The evaluation questionnaire to be gathered will be analyze using descriptive statistics. The verbal description is based on the following ranges of value reflected in Table 2.

Table 2. Evaluate System's Usability and Functionality

NUMERICAL VALUE	DESCRIPTION	VALUE
1	Strongly Disagree	1.00-1.49
2	Disagree	1.50-2.49
3	Agree	2.50-3.49
4	Strongly Disagree	3.50-4.00

Evaluation Measure

In In the last phase of the study, a confusion matrix for many classes will be used to assess the algorithm's accuracy rate.

According to Mohajon (2020), making a confusion matrix is a simple way to assess how well your prediction model is working. Each row in a confusion matrix, per his paper, represents the frequency of predictions the model made while classifying the classes correctly or erroneously. In the confusion, the topics from Chapter 1, Chapter 2, Chapter 3 will be the value for True Positive (TP) and the topics from Chapter 4 and Chapter 5 will be the value for True Negative (TN). matrix for numerous classes, with the negative value corresponding. Six groups will be created from the confusion matrix's result. The following are the Four categories:

a. **True Positive (TP)** - if the file topics from Chapter 1, Chapter 2, Chapter 3 are classified as topics from Chapter 1, Chapter 2, Chapter 3.

b. **True Negative (TN)** - if the file topics from Chapter 4 and Chapter is classified as topics from Chapter 4 and Chapter 5

c. **False Positive (FP)** - if the file topics from Chapter 1, Chapter 2, Chapter 3 is classified as topics from Chapter 4 and Chapter 5.

d. **False Negative (FN)** - if the file topics from Chapter 4 and Chapter 5 is classified as topics from Chapter 1, Chapter 2, Chapter 3.

It is important to evaluate the accuracy, precision, and recall of the suggested framework. A machine learning model's evaluation metric should normally be the confusion matrix. It provides you with a straightforward yet precise performance evaluation for your model. The confusion matrix's most popular performance metrics are listed below:

Accuracy - It provides you with the model's overall accuracy, or the percentage of all samples that the classifier identified correctly. The correctness should be evaluated using the following equation:

$$(\mathrm{TP} + \mathrm{TN})/(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}) \tag{1}$$

Precision - It displays the percentage of forecasts in the "positive" class that were true "positives." To calculate precision, use the formula below:

$$\mathrm{TP}/(\mathrm{TP} + \mathrm{FP}) \tag{2}$$

Recall - It shows the classifier's percentage of all positive samples. Accurately discovered and found to be favorable. It is also known as sensitivity, true positive rate (TPR), and chance of detection. Recall should be calculated using the formula below:

$$\mathrm{TP}/(\mathrm{TP} + \mathrm{FN}) \tag{3}$$

Testing and Documentation

he testing and research results to evaluate the algorithm's accuracy base on training and testing file datasets are displayed in Figure 3. A confusion matrix table is used in Figure 3 to show the documentation and testing results sample.

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

Figure 3. Confusion Matrix Sample

Accuracy: TP+TN / TP+TN+FP+FN

Precision: TP / TP+TN

Recall: TP / TP + FP

System Requirements Specification

This section shows the software and hardware system requirements specification.

Hardware Specifications

Table 3 shows the hardware specification to be used in the creation of the framework. It is required for the framework to run efficiently.

Table 3. Hardware Specification Table

HARDWARE	SPECIFICATION
Processor	Core i5
Memory	8GB
Solid State Drive	256GB

Software Specifications

Table 4 shows the software specification to be used in the creation of the framework. It is required for the framework to run efficiently.

Table 4. Software Specification Table

SOFTWARE	SPECIFICATION
Processor	Windows 10
Programming Language	Python
IDE Library	Visual Studio Code

User Interface of Prototype

This section shows the user interface design of perceived prototype of the web application for multiple-choice question quiz maker. Figure 4 represents the homepage or main interface design for the web application used to create multiple-choice question quizzes. The upload arrow symbol indicates where users will upload the PDF or Word document file to be utilized in the creation of the quiz.

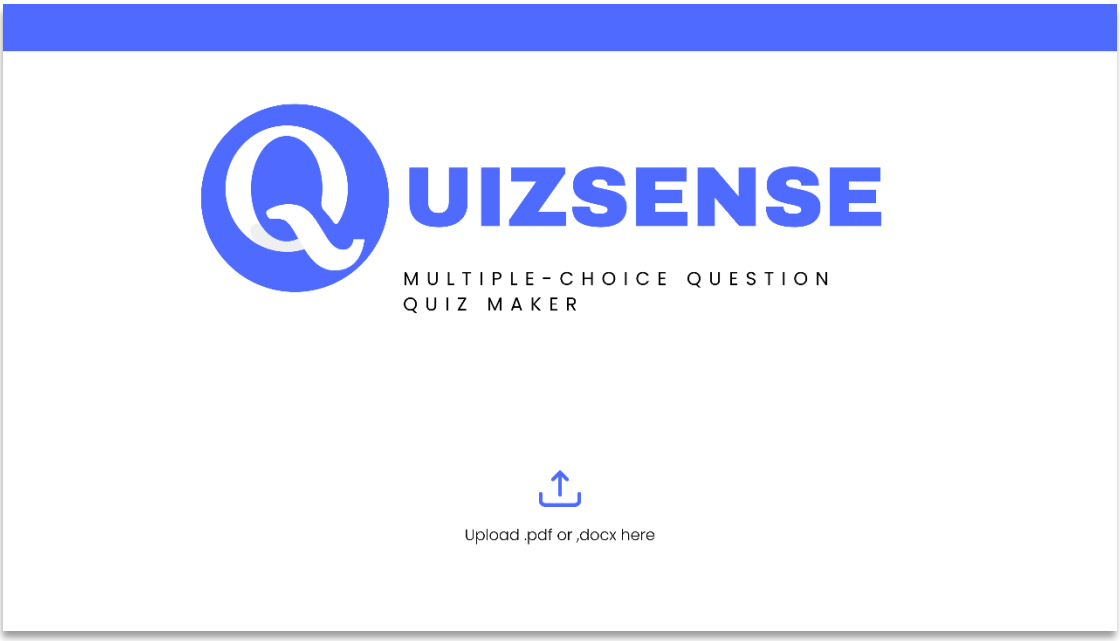

Figure 4. Quiz Maker Landing Page

Figure 5 displays a quiz interface from the web application. It showcases a multiple-choice question as part of a generated quiz. The radio button is where the user will provide their answer.

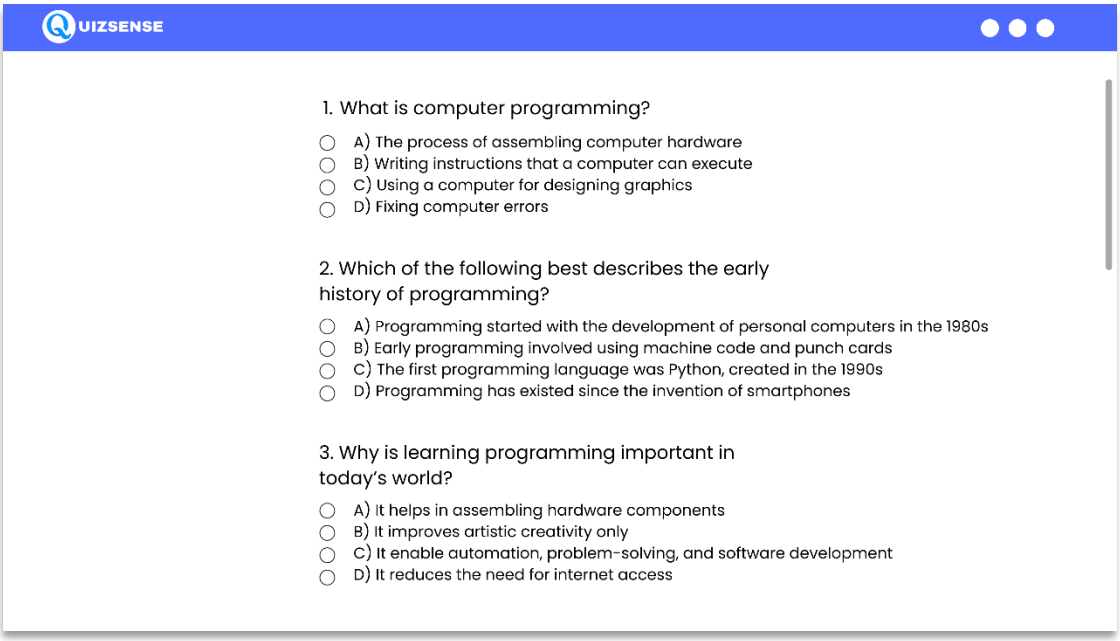

Figure 5. Multiple-Choice Question Quiz Interface

Figure 6 shows summarizing the user performance on a multiple-choice quiz.

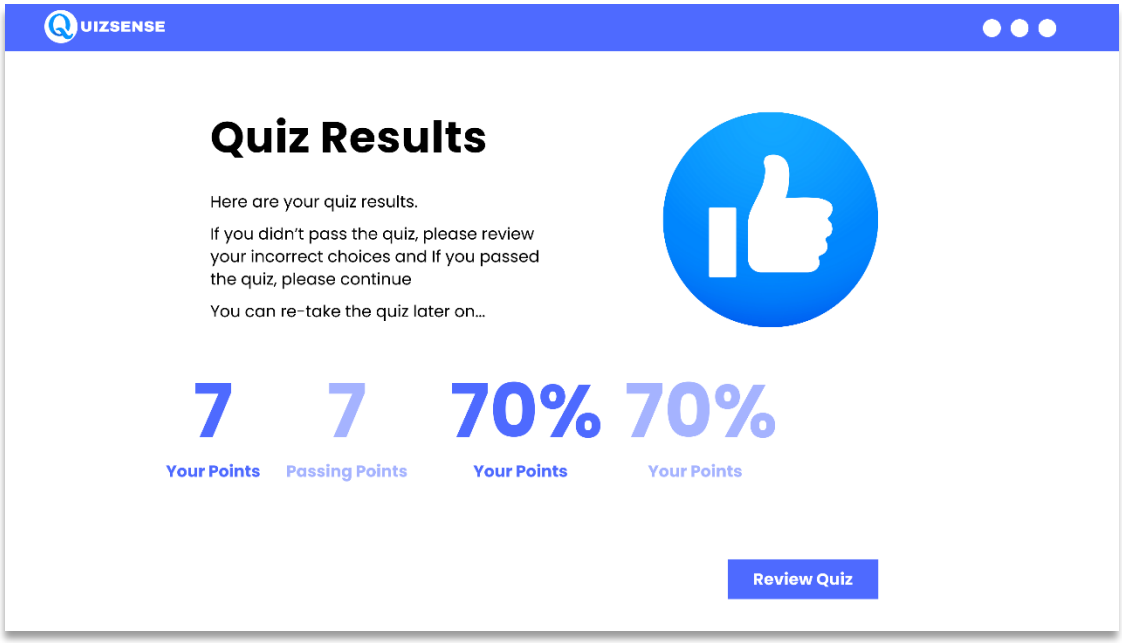

Figure 6. Quiz Results Page

Figure 7 displays a completed quiz with answers provided for each question.

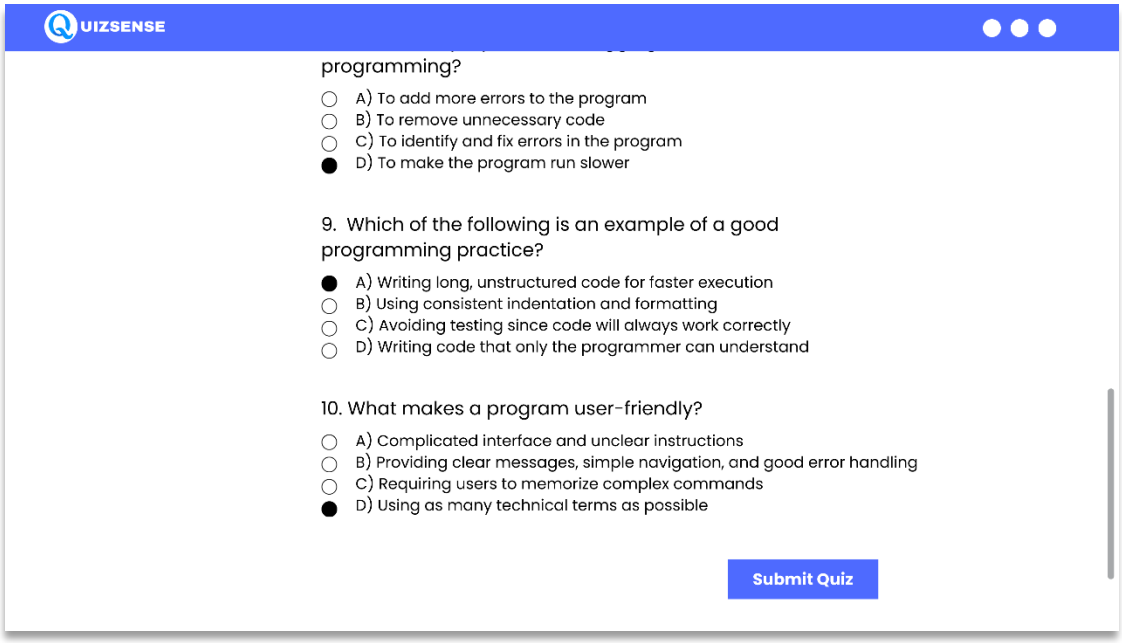

Figure 7. Review Quiz Page

Figure 8 shows feedback to improve topic areas based on the user's weaknesses identified through quiz performance.

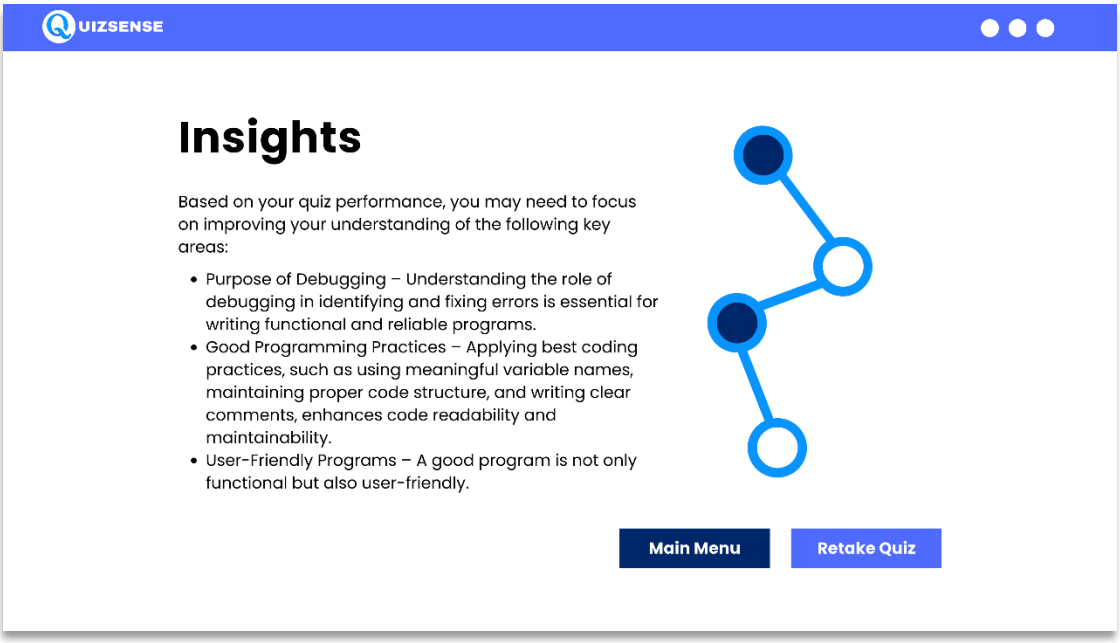

Figure 8. Insights Page

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APPENDICES

Appendix A. Application for Research Adviser

UNIVERSITY OF SOUTHERN MINDANAO
Kabacan, Cotabato
Philippines

APPLICATION FOR RESEARCH ADVISER

Date: 2024. 12. 13

ELIZABETH R. GENOTIVA
Department of Computing and Library Information Science
College of Engineering and Information Technology
USM, Kabacan, Cotabato

Ma'am:

I would like to request that you will be my Research adviser effective 1st semester, SY 2024-2025. I intend to work on **"QUIZSENSE: A WEB APPLICATION FOR MULTIPLE-CHOICE QUESTION QUIZ MAKER."**

I am hoping for your most favorable approval on this request. Thank you very much.

Very truly yours,

ARIFEH OSAMA C. CAMANTO
Printed Name and Signature of Student

APPROVED

ELIZABETH R. GENOTIVA
Adviser

2024. 12. 13
Date

USM-EDR-F01-Rev.4.2022.10.18

Appendix B. Application for Research Title

UNIVERSITY OF SOUTHERN MINDANAO
Kabacan, Cotabato
Philippines

APPLICATION FOR RESEARCH TITLE

Date: 2024. 12. 13

ARJAY S. AGBUNAG
Chairperson, Department of Computing and Library Information Science

SIR:

I would like to request your office to allow me to research on the study entitled **"QUIZSENSE: A WEB APPLICATION FOR MULTIPLE-CHOICE QUESTION QUIZ MAKER."**

The study has the following objectives:

1. gather pdf or word file of fundamentals of programming as dataset;
2. evaluate the accuracy of the web application using confusion matrix;
3. develop a web application that allows students to take a quiz;
4. recommend topics improvement based on quiz performance; and
5. evaluate the usability and functionality of the web application.

Very truly yours,

ARIFEH OSAMA C. CAMANTO
Printed Name and Signature of Student

NOTED	
ELIZABETH R. GENOTIVA Adviser	2024. 12. 13 Date
NOR-AINE M. CORPUZ Department Research Coordinator	2024. 12. 13 Date
SHIERYL PALOMA ORTIZA College Research Coordinator	2024. 12. 13 Date
APPROVED	
ARJAY S. AGBUNAG Department Chairperson	2024. 12. 13 Date

USM-EDR-F02-Rev.4.2021.11.04

Appendix C. Estimated Budget of the Research

UNIVERSITY OF SOUTHERN MINDANAO
Kabacan, Cotabato
Philippines

ESTIMATED BUDGET OF THE RESEARCH

Title of Study
QUIZSENSE: A WEB APPLICATION FOR MULTIPLE-CHOICE QUESTION QUIZ MAKER

ITEMS/DESCRIPTION	ESTIMATED COST
A. BOND PAPER	450.00
PRINTING	350.00
CLIPS	100.00
B. FOOD	800.00
TRAVEL	300.00
Grand Total	2,000.00

Prepared and submitted by:

ARIFEH OSAMA C. CAMANTO
Printed Name and Signature of the Student

NOTED	
ELIZABETH R. GENOTIVA Adviser	2024. 12. 20 Date
NOR-AINE M. CORPUZ Department Research Coordinator	2024. 12. 20 Date
ARJAY S. AGBUNAG Department Chairperson	2024. 12. 20 Date

Appendix D. Application for Thesis Outline Defense

UNIVERSITY OF SOUTHERN MINDANAO
Kabacan, Cotabato
Philippines

APPLICATION FOR THESIS OUTLINE DEFENSE

Name	**ARIFEH OSAMA C. CAMANTO**
Degree/Major	**BACHELOR OF SCIENCE IN COMPUTER SCIENCE**
Thesis Title	**QUIZSENSE: A WEB APPLICATION FOR MULTIPLE-CHOICE QUESTION QUIZ MAKER**
Date of Examination	**DECEMBER 18, 2024**
Time	**1:00 PM**
Place	**COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY, COMPUTING AND LIBRARY INFORMATION SCIENCE OFFICE**

MEMBERS OF THE EXAMINING COMMITTEE

Name	Signature	Date
DANILYN A. FLORES		2024. 12. 15
ARJAY S. AGBUNAG		2024. 12. 15
JOSEPH C. LORILLA		2024. 12. 15

RECOMMENDING APPROVAL:

ELIZABETH R. GENOTIVA
Adviser

Co-Adviser (Optional)

APPROVED:

NOR-AINE M. CORPUZ
Department Chairperson

College Statistician
(Optional)

ARJAY S. AGBUNAG
Department Research Coordinator

REPORT ON THE RESULT OF EXAMINATION

Name	Signature	Remarks
DANILYN A. FLORES		PASSED
ARJAY S. AGBUNAG		PASSED
JOSEPH C. LORILLA		passed

APPROVED:

ARJAY S. AGBUNAG
Department Research Coordinator
2024. 12. 19
Date

USM-EDR-F07-Rev.3.2020.02.24

UNIVERSITY OF SOUTHERN MINDANAO
Kabacan, Cotabato
Philippines

CURRICULUM VITAE

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PERSONAL INFORMATION	
Last Name	**CAMANTO**
First Name	**ARIFEH OSAMA**
Middle Name	**CAMSA**
Nickname	**DATU**
Age	**22**
Nationality	**FILIPINO**
Religion	**ISLAM**
Civil Status	**SINGLE**
Father's Name	**ADPAIZAR MANGKIAO CAMANTO**
Mother's Name	**HYDEE CAMSA CAMANATO**
Educational Background	
Elementary	**KABACAN PILOT CENTRAL SCHOOL 2008 – 2014**
Junior High School	**UNIVERSITY OF SOUTHERN MINDANAO – UNIVERSITY LABORATORY SCHOOL 2014-2018**
Senior High School	**UNIVERSITY OF SOUTHERN MINDANAO – UNIVERSITY LABORATORY SCHOOL 2018-2020**
Tertiary	**UNIVERSITY OF SOUTHERN MINDANAO - MAIN CAMPUS**

	2020 – 2025
TRAINING AND PROFESSIONAL DEVELOPMENT	
IT Support, On-The-Job Training (OJT)	**AWESOME OS, Davao City** *July 2024 – August 2024*
MEMBERSHIP TO ORGANIZATIONS/SOCIETIES	
University Performing Arts Guild – Sibol Films	**Cinematographer** *January 2023 – November 2023*
College of Engineering and Information Technology – Local Student Government	**Editor** *August 2022 – June 2023* **Treasurer** *August 2023 – June 2024*
Philippine Society of Information Technology Students	**Member** *August 2021 – Present*